IRF530

June 24, 2026

Carga máxima de Gate 26 nC Carga gate-dreno 11 nC Carga gate-source 5.5 nC Carga dreno-source 11 nC

Tensão de Treshold 4V Transcondutância direta 5.1S Capacitância de entrada 670 pF Capacitância de saída 250 pF Rise Time 34 ns fall time 24 ns resistência interna 1 ohm

[1]:
```
Qg_max = 26e-9 #Maximum gate charge
Qgd = 11e-9 #Gate-drain charge
Qgs = 5.5e-9 #Gate-source charge
Qds = 11e-9 #Drain-source charge
Vth = 4 #Threshold voltage
Gfs = 5.1 #Forward transconductance
Ciss = 670e-12 #Input capacitance
Coss = 250e-12 #Output capacitance
Crss = 60e-12 #Reverse capacitance
tr = 34e-9 #Rise time
tf = 24e-9 #Fall time
Rint = 1 #Internal resistance
fs = 40e3
```

[2]:
```
VG = 15 # Tensão de acionamento
ID = 1.25 # Corrente de dreno
VDS = 30 # Tensão de saída de dreno
D = 0.4 # Razão cíclica
RL = 1e3 # Reistência de carga
VTTL = 3.3 # Tensão de controle TTL
```

[3]:
```
print("Correntes de acionamento")

print("Considerando Carga máxima de gate:")
Ig_on = Qg_max/tr
print(f"Gate current during turn-on: {Ig_on:.2e} A")
Ig_off = Qg_max/tf
print(f"Gate current during turn-off: {Ig_off:.2e} A")


print("Considerando Qgs1 + Qgd:")
Ig_on2 = (Qgd + Qgs)/tr
```

```
print(f"Gate current during turn-on: {Ig_on2:.2e} A")
Ig_off2 = (Qgd + Qgs)/tf
print(f"Gate current during turn-off: {Ig_off2:.2e} A")
```

```
Correntes de acionamento
Considerando Carga máxima de gate:
Gate current during turn-on: 7.65e-01 A
Gate current during turn-off: 1.08e+00 A
Considerando Qgs1 + Qgd:
Gate current during turn-on: 4.85e-01 A
Gate current during turn-off: 6.88e-01 A
```

[4]:
```
print("Calculo resistor de Gate")

print("Considerando Carga máxima de gate:")
Ron1 = VG/Ig_on
print(f"Gate resistor for turn-on: {Ron1:.2f} Ohms")
Roff1 = VG/Ig_off


print("Considerando Qgs1 + Qgd:")
Ron2 = VG/Ig_on2
print(f"Gate resistor for turn-on: {Ron2:.2f} Ohms")
Roff2 = VG/Ig_off2
print(f"Gate resistor for turn-off: {Roff2:.2f} Ohms")
```

```
Calculo resistor de Gate
Considerando Carga máxima de gate:
Gate resistor for turn-on: 19.62 Ohms
Considerando Qgs1 + Qgd:
Gate resistor for turn-on: 30.91 Ohms
Gate resistor for turn-off: 21.82 Ohms
```

[5]:
```
import numpy as np

print("Tempos de comutação")
td_on = (Ron2 + Rint) * Ciss * np.log((VG) / (VG - Vth ))
print(f"Turn-on delay time: {td_on:.2e} s")

tri = (Ron2 + Rint) * Ciss * np.log((Gfs * (VG - Vth)/(Gfs * (VG - Vth) - ID)) )
print(f"Turn-off delay time: {tri:.2e} s")

VGPL = Vth + (ID/Gfs)
VDS_on = 0
print(f"Gate voltage for full conduction: {VGPL:.2f} V")

tfv = (Ron2 + Rint) * Ciss * ((VDS - VDS_on)/(VG - VGPL))
```

```
print(f"Fall time: {tfv:.2e} s")

Pon = 0.5 * VDS * ID * (tri + tfv) * fs
print(f"Switching power (turn-on/off): {Pon:.2e} W")
```

```
Tempos de comutação
Turn-on delay time: 6.63e-09 s
Turn-off delay time: 4.82e-10 s
Gate voltage for full conduction: 4.25 V
Fall time: 5.96e-08 s
Switching power (turn-on/off): 4.51e-02 W
```

[6]:
```
print("Circuito Totem Pole")

Ib2 = Ig_on2/10
print(f"Base current for the second transistor: {Ib2:.2e} A")
IC1 = Ib2

Rc = VG/IC1
print(f"Collector resistor for the first transistor: {Rc:.2f} Ohms")

Ib1 = IC1/10
print(f"Base current for the first transistor: {Ib1:.2e} A")

RB = (VTTL - 0.7)/Ib1
print(f"Base resistor for the first transistor: {RB:.2f} Ohms")
```

```
Circuito Totem Pole
Base current for the second transistor: 4.85e-02 A
Collector resistor for the first transistor: 309.09 Ohms
Base current for the first transistor: 4.85e-03 A
Base resistor for the first transistor: 535.76 Ohms
```

[7]:
```
print("IR2110")
Vbo = 15 # Tensão da fonte de boostrap
Vgs_min = 14 # Tensão mínima de gate para garantir a condução total
dead_time = 650e-9 # Dead Time
Qsw = Qgs + Qgd # Carga eletrica para acionamento
Igss = 100e-9 # Corrente de fuga entre gate e source
IQBS = 100e-6 # Corrente de quiescente entre pinos VB e VS
Ilk = 50e-6 # Corrente de fuga do driver
delta_Vbo = 1 # Variação da tensão do capacitor de bootstrap
Vfb = 0.7 # Queda de tensão no diodo de bootstrap
```

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IR2110
```

[8]:
```python
Cg = Qsw/(Vbo - Vfb)
print(f"Capacitor de Gate: {Cg:.2e} F")

Cbo = 10 * Cg
print(f"Capacitor de Bootstrap: {Cbo:.2e} F")

Vboot = Vbo - Vfb - Vgs_min
print(f"Máxima queda de tensão no capacitor de bootstrap: {Vboot:.2f} V")

tmax = D/fs
print(f"Tempo máximo para descarga do capacitor de bootstrap: {tmax:.2e} s")

QT = Qsw + (IQBS + Ilk + Igss) * tmax
print(f"Carga máxima que o capacitor deve fornecer no acionamento: {QT:.2e} C")

Cboot = QT/delta_Vbo
print(f"Capacitância mínima do capacitor de bootstrap: {Cboot:.2e} F")
```

```
Capacitor de Gate: 1.15e-09 F
Capacitor de Bootstrap: 1.15e-08 F
Máxima queda de tensão no capacitor de bootstrap: 0.30 V
Tempo máximo para descarga do capacitor de bootstrap: 1.00e-05 s
Carga máxima que o capacitor deve fornecer no acionamento: 1.80e-08 C
Capacitância mínima do capacitor de bootstrap: 1.80e-08 F
```